A Mathematical Discussion of Quantum Tunneling

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Introduction

In this article, I demonstrate how quantum particles with energy E less than a step-potential barrier V_0 starting at $x = 0$ (terminating at $x = a$) can be found in the classically disallowed region of $x \geq a$.

Although quantum tunnelling initially seems like a very theoretical concept, one soon realises that it is essential to many processes that keep us humans alive. The temperature in stellar cores is generally insufficient to impart enough energy to nuclei to overcome the Coulomb repulsion barrier. Quantum tunnelling increases the probability of penetrating this barrier, enabling the fusion reaction to be sustained. This is ***a little*** important; without the Sun you wouldn't be reading this article.

Please note that I don't have much knowledge of nuclear physics and stellar nucleosynthesis, so take my calculations and discussion in Section 2 with a grain of salt.

As always, comments, queries, and suggestions are most welcome:

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Do also email me should you find an error (see italicised text above).

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1 The Potential Barrier

Consider a one-dimensional, time-independent potential defined by

$$V(x) = \begin{cases} V_0 & ,0 < x < a \\ 0, & ,\text{otherwise.} \end{cases}$$

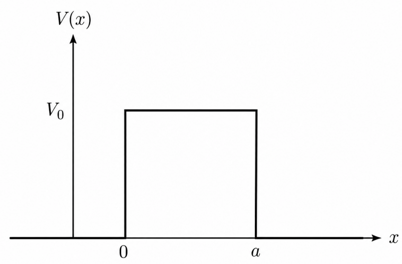

Figure 1: Plot of the potential $V(x)$. Image generated by ChatGPT 5.6 Luna

Classically speaking, if a particle of mass m moving to the right has energy less than the potential, $E < V_0$, reflection would occur since penetrating the barrier would mean the particle would have a negative kinetic energy. Quantum mechanically, however, this is not the case. The main perpetrator behind this discrepancy is the time-independent Schrödinger Equation, which in one dimension is given by

$$-\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} + V(x)\psi(x) = E\psi(x).$$

By definition of our potential, the Schrödinger Equations in the regions $x \leq 0$, $0 < x < a$ and $a \leq x$ are

$$-\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} = E\psi(x),$$

$$-\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} + V_0\psi(x) = E\psi(x),$$

and

$$-\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} = E\psi(x),$$

respectively.

Defining the quantities $k_1 = \sqrt{2mE/\hbar^2}$ and $k_2 = \sqrt{2m(E - V_0)/\hbar^2}$, and rearranging the three equations above yields the familiar harmonic differential equations

$$\begin{aligned}
\frac{d^2\psi}{dx^2} + k_1^2\psi = 0 \\
\frac{d^2\psi}{dx^2} + k_2^2\psi = 0 \\
\frac{d^2\psi}{dx^2} + k_1^2\psi = 0
\end{aligned}$$

in the regions $x \leq 0$, $0 < x < a$, and $a \leq x$.

The solutions to each of these equations are linear combinations of the exponentials e^{ik_jx}, and e^{-ik_jx}, where $j = 1, 2$. Therefore, we may write the overall wave function as

$$\psi(x) = \begin{cases} \psi_1(x), & x \leq 0, \\ \psi_2(x), & 0 < x < a, \\ \psi_3(x), & a \leq x, \end{cases}$$

where the ψ_k are the solutions to the differential equations above:

$$\begin{aligned}
\psi_1(x) &= A_1 e^{ik_1x} + A_2 e^{-ik_1x} \\
\psi_2(x) &= B_1 e^{ik_2x} + B_2 e^{-ik_2x} \\
\psi_3(x) &= C_1 e^{ik_1x} + C_2 e^{-ik_1x}.
\end{aligned}$$

However, note that the complex exponential e^{-ik_1x} corresponds to a wave propagating to the left, which cannot happen in the region $x \geq a$ (there is no barrier for reflection). This forces $C_2 = 0$, so that

$$\begin{aligned}
\psi_1(x) &= A_1 e^{ik_1x} + A_2 e^{-ik_1x} \\
\psi_2(x) &= B_1 e^{ik_2x} + B_2 e^{-ik_2x} \\
\psi_3(x) &= C_1 e^{ik_1x}.
\end{aligned}$$

The transmission coefficient T of the wave function is defined by

$$T = \left|\frac{J_t}{J_i}\right|$$

where J_t and J_i represent the transmitted and incident current densities, respectively.

The incident waves are given by $\psi_i(x) = A_1 e^{ik_1 x}$ and the transmitted waves are given by $\psi_t(x) = \psi_3(x) = C_1 e^{ik_1 x}$. Now, recall

$$\begin{aligned}
J_i &= \frac{i\hbar}{2m}(\psi_i \nabla \psi_i^* - \psi_i^* \nabla \psi_i) \\
&= \frac{i\hbar}{2m}\left(\psi_i(x)\frac{d\psi_i^*}{dx} - \psi_i^*(x)\frac{d\psi_i}{dx}\right) \\
&= \frac{i\hbar}{2m}\left(A_1 e^{ik_1 x}(-ik_1 A_1^* e^{-ik_1 x}) - A_1^* e^{-ik_1 x}(ik_1 A_1 e^{ik_1 x})\right) \\
&= \frac{\hbar k_1}{m}|A_1|^2.
\end{aligned}$$

Similarly, the transmitted current density is given by

$$J_t = \frac{\hbar k_1}{m}|C_1|^2,$$

and it follows that the transmission coefficient[1] is given by

$$T = \frac{|C_1|^2}{|A_1|^2}.$$

To determine the constants A_2, B_1, B_2 and C_1 in terms of A_1, we must invoke boundary conditions on ψ. Recall that the wave function ψ (along with its first derivative) must each be continuous at $x = 0$ and $x = a$:

$$\begin{aligned}
\psi_1(0) &= \psi_2(0) & \qquad \psi_1'(0) &= \psi_2'(0) \\
\psi_2(a) &= \psi_3(a) & \qquad \psi_2'(0) &= \psi_3'(0).
\end{aligned}$$

Some algebra (see page 255) of [Zet22]) shows that

$$T = \frac{|C_1|^2}{|A_1|^2} = 4\left[4\cosh^2(k_2 a) + \left(\frac{k_2^2 - k_2^2}{k_1 k_2}\right)^2 \sinh^2(k_2 a)\right]^{-1}.$$

Using the hyperbolic identity $\cosh^2 x = 1 + \sinh^2 x$, as can be found in the front cover of [Tay05], we may simplify the expression above to

[1] We do not bother calculating the reflection coefficient R, but one can verify that $R+T = 1$, using $R = |J_i/J_r|$.

$$T = \left[1 + \frac{1}{4}\left(\frac{k_1^2 + k_2^2}{k_1 k_2}\right)^2 \sinh^2(k_2 a)\right]^{-1}.$$

It should be noted that T is *non-zero*, which implies that the probability of transmission of the particles into the region $x \geq a$ is not zero, which is impossible in classical physics. This is quantum tunnelling. For further discussion on potential barriers (and wells), see [Zet22].

2 Quantum Tunneling in Alpha Decay

Despite the supposed absurdity of the conclusion of the previous section, quantum tunnelling is a *real phenomenon.* One way that quantum tunnelling was observed was through α-particle decay, such as that with the uranium-238 isotope $^{238}_{92}\mathrm{U}$. The nuclear equation for this α-emission is

$$^{238}_{92}\mathrm{U} \longrightarrow {}^{234}_{90}\mathrm{Th} + {}^{4}_{2}\mathrm{He}$$

We can calculate the energy released in this decay using Einstein's mass-energy equivalence ($E = mc^2$) and the masses of each isotope:

Isotope	Mass/amu
$^{238}_{92}\mathrm{U}$	238.0507882
$^{234}_{90}\mathrm{Th}$	234.043601
$^{4}_{2}\mathrm{He}$	4.00260325413

Table 1: Isotope masses for the species in the decay above

Now, the Q value for this reaction is

$$\begin{aligned} Q &= \Delta m c^2 \\ &= (m_i - m_f)c^2 \\ &= (238.0507882\mathrm{u} - (4.00260325413\mathrm{u} + 234.043601\mathrm{u}))(3.00 \times 10^8 \mathrm{m\ s}^{-1}) \\ &\approx +4.27\mathrm{MeV}. \end{aligned}$$

It follows that the decay process is

$$^{238}_{92}\mathrm{U} \longrightarrow {}^{234}_{90}\mathrm{Th} + {}^{4}_{2}\mathrm{He} + 4.27\,\mathrm{MeV}.$$

Now, the Coulomb barrier of the uranium-238 nucleus is roughly 25 MeV, while the decay process releases only 4.27 MeV. Classically speaking, therefore, an alpha particle can't be emitted. Therefore, the only plausible explanation is that the alpha particle strikes the potential barrier, with each 'collision'[2] carrying a small probability of it tunnelling through. This is the quantum tunnelling theory of alpha decay, developed by George Gamow [Gam28] and by Ronald Wilfred Gurney and Edward Condon [GC28] independently in 1928.

References

[Gam28] George Gamow. "Zur Quantentheorie des Atomkernes". In: *Zeitschrift für Physik* 51.3–4 (1928), pp. 204–212. DOI: `10.1007/BF01343196`.

[GC28] Ronald W. Gurney and Edward U. Condon. "Wave Mechanics and Radioactive Disintegration". In: *Nature* 122.3073 (1928), p. 439. DOI: `10.1038/122439a0`.

[Tay05] John R. Taylor. *Classical Mechanics*. Sausalito, CA: University Science Books, 2005. ISBN: 978-1-891389-22-1.

[Zet22] Nouredine Zettili. *Quantum Mechanics*. Wiley, 2022. ISBN: 978-1-118-30789-2.

[2] The process is actually quite extreme, with the alpha particle colliding with the barrier 10^{21} times per second.